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Title: Endangered Plant and Animal Species of Maharashtra: A Study on Biodiversity Loss

Aim: To collect the information on endangered plant and animal species from Maharashtra, document their taxonomic classification, analyze the anthropogenic causes of their population decline, and arrive at a conclusion based on the analysis of data.

Project Writing

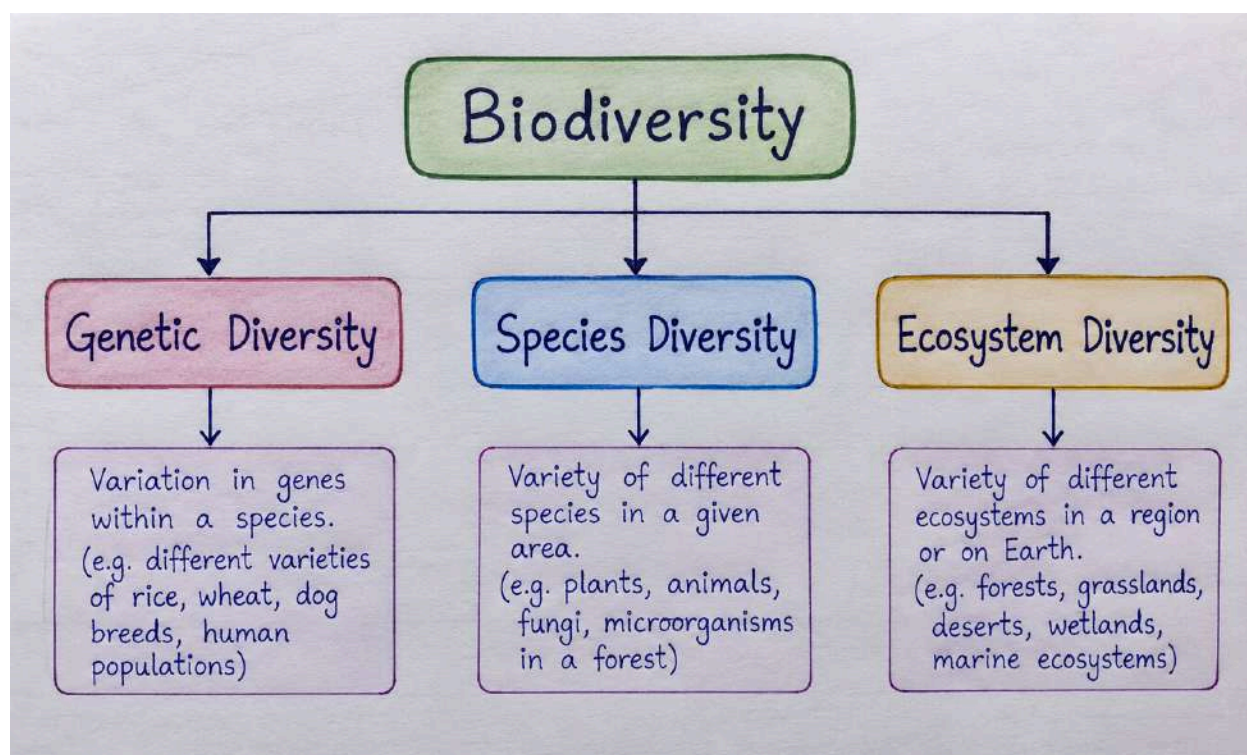
1. Name of the student and concerned details

- **Name:** Jayant Manoj Deore
- **Standard:** 12th (Science)
- **Subject:** Biology
- **Project Topic:** 3 (Collect the information on endangered plant and animal species from Maharashtra)
- **Academic Year:** 2026-2027

2. Acknowledgements

I would like to express my sincere gratitude to my Biology teacher for their invaluable guidance in planning the approach and methodology for this project. Their continuous support helped me understand the ecological concepts clearly. I am also deeply thankful to our Principal for providing the necessary academic environment. Lastly, I extend my heartfelt thanks to my parents and classmates for their encouragement and assistance in compiling this data.

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3. Table of contents:

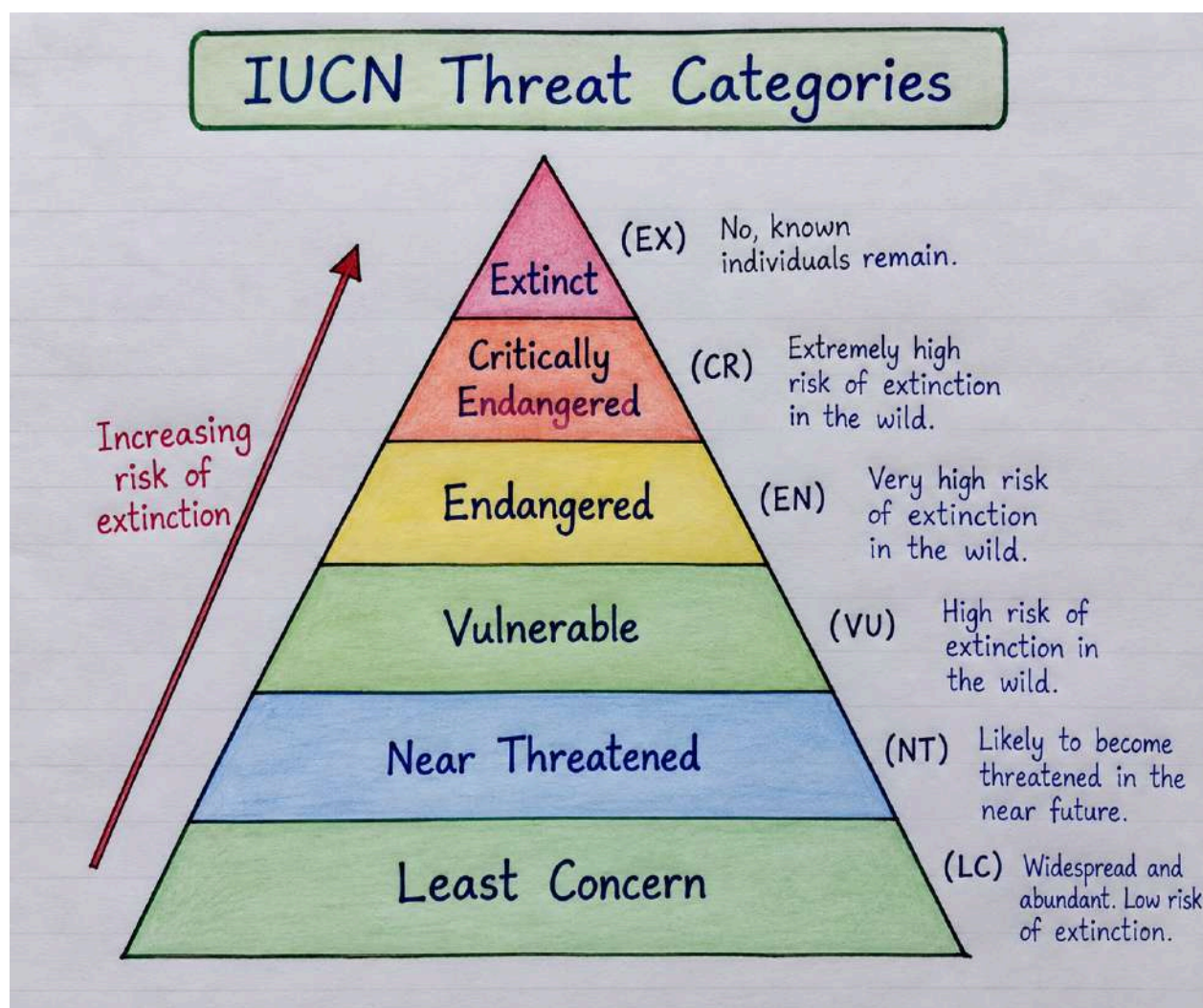
1. Title and Aim
2. Name of the student and concerned details
3. Acknowledgements
4. Table of contents
5. Actual Report
 - a. Introduction of the topic and need for project
 - b. Description of method/s used
 - c. Data collected (Endangered Fauna & Flora)
 - d. Analysis of collected data
 - e. Conclusion
6. Bibliography

4. Actual Report:

a. Introduction of the topic and need for the project Introduction to Biodiversity:

Biodiversity refers to the totality of genes, species, and ecosystems in a given region. The state of Maharashtra is endowed with immense ecological richness, particularly in the Western Ghats (Sahyadris), which is globally recognized as a "**Biodiversity Hotspot**". This region exhibits a very high degree of endemism, meaning a large percentage of the plant and animal species found here exist nowhere else in the entire world.

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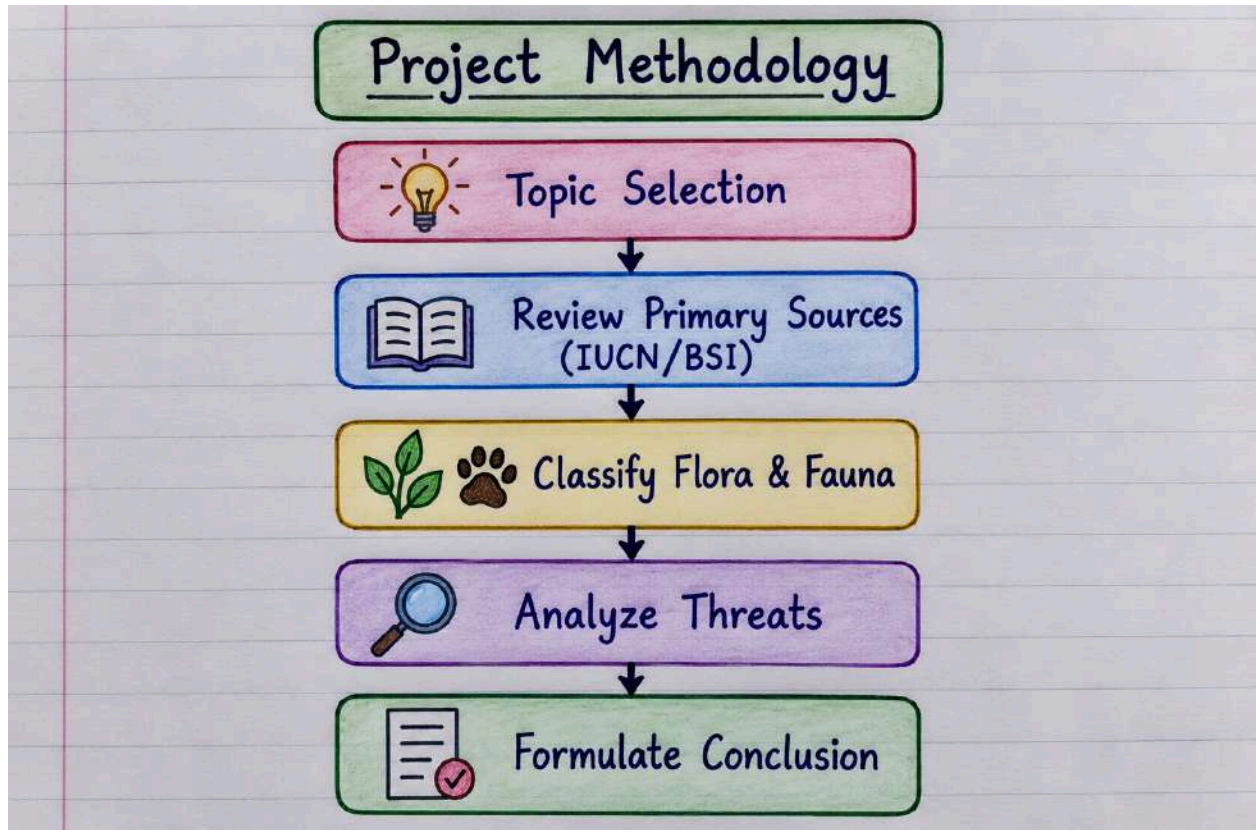
According to the **International Union for Conservation of Nature (IUCN)**, species are systematically categorized based on their extinction risk. For the scope of this project, we are focusing on species classified as "**Endangered**" (EN), which face a very high risk of extinction in the wild, and "**Critically Endangered**" (CR), which face an extremely high risk of extinction in the immediate future.

Need for the Project: Due to rapid urbanization, unchecked industrialization, and massive deforestation, Maharashtra's natural habitats are shrinking at an alarming rate. This project is highly necessary to systematically document the specific native flora and fauna that are currently on the verge of extinction. Understanding their unique ecological roles and precisely identifying the anthropogenic (human-made) threats they face is the crucial first step toward effective conservation and creating environmental awareness among the youth.

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(Action: Draw a neat flowchart titled "Project Planning & Methodology")

- **Step 1:** Select the topic based on feasibility.
- **Step 2:** Define the objectives of the project.
- **Step 3:** Use primary sources for information-based data collection.
- **Step 4:** Formulate the data gathered to aid in analysis.
- **Step 5:** Arrive at a conclusion based on the analysis.



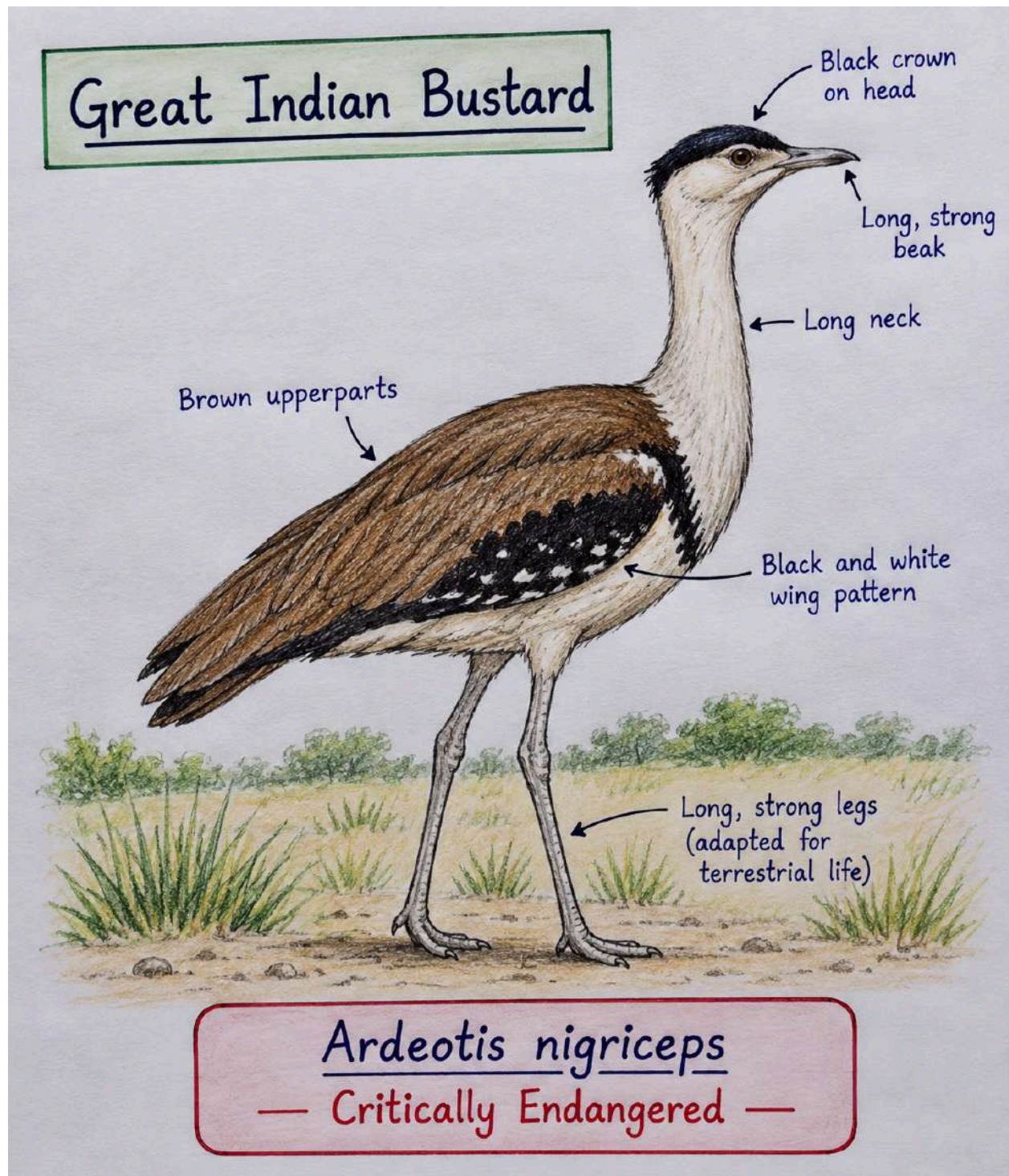
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b. Description of method/s used As per the strict project guidelines for this topic, the approach and methodology rely on primary sources for an information-based project. Direct experimentation for collection or study of processes on these species is prohibited due to their highly protected legal status under the Wildlife Protection Act (1972).

- **Primary Sources:** Secondary data was meticulously collected from highly reliable, peer-reviewed scientific sources. The global threat status of each selected species was verified using the official IUCN Red List of Threatened Species.
- **Regional Data Formulation:** Specific habitat details and localized distribution within Maharashtra were gathered using various techniques so as to aid in analysis. Information was referenced from the Botanical Survey of India (BSI) and the Zoological Survey of India (ZSI).
- **Data Organization:** The information was systematically categorized into Fauna (Animals) and Flora (Plants), detailing their taxonomic classification and primary threats.

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(Action: Print and paste the image file 7724.png here. It highlights the exact morphological features of the Great Indian Bustard).



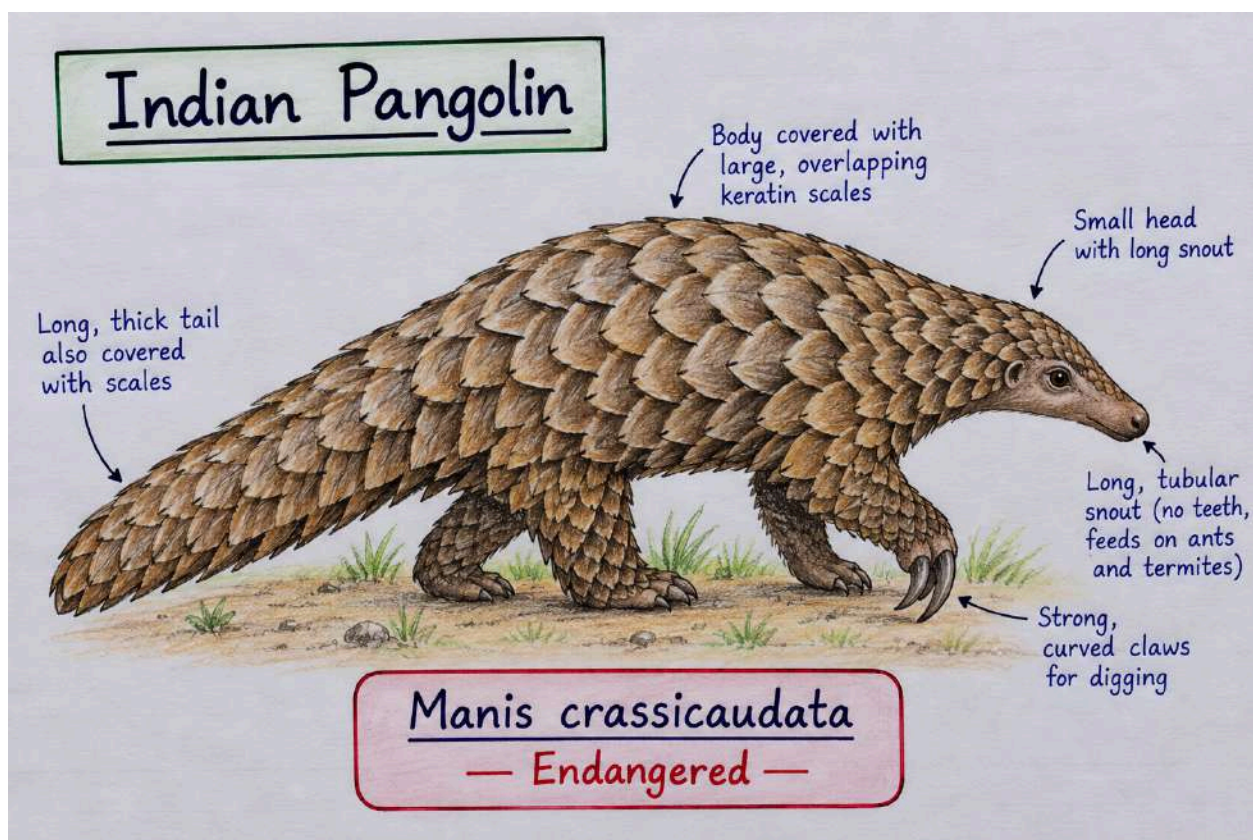
Part I: Endangered Fauna (Animals) of Maharashtra

1. Great Indian Bustard

- **Taxonomic Classification:**
 - **Kingdom: Animalia**
 - **Phylum: Chordata**
 - **Class: Aves**
 - **Order: Otidiformes**
 - **Scientific Name: *Ardeotis nigriceps***
- **IUCN Status:** Critically Endangered (CR).
- **Habitat in Maharashtra:** Dry, semi-arid grasslands of Nanaj (Solapur) and Ahmednagar.
- **Characteristics:** A large, heavy bird with an ostrich-like appearance, a black crown on its head, and long strong legs adapted for terrestrial life. It acts as an umbrella species for healthy grassland ecosystems.
- **Primary Threats:** The leading cause of mortality is fatal collisions with high-tension electric power lines crossing their flight paths. Furthermore, habitat loss due to the conversion of natural grasslands into intensive agricultural lands destroys their ground nests.

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(Action: Print and paste the image file **7725.png** here. It showcases the Indian Pangolin's overlapping keratin scales and strong claws).



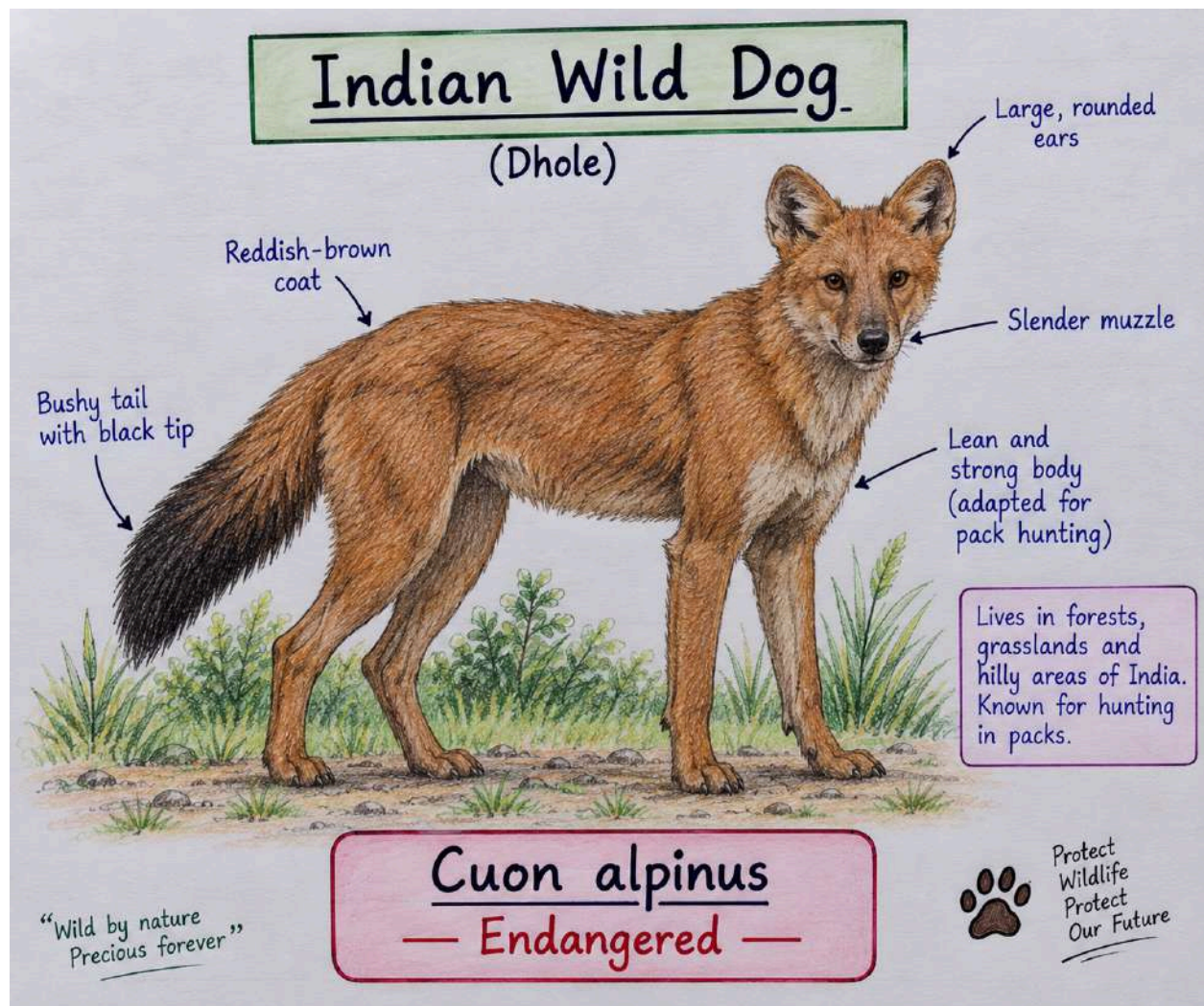
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2. Indian Pangolin

- **Taxonomic Classification:**
 - **Kingdom: Animalia**
 - **Phylum: Chordata**
 - **Class: Mammalia**
 - **Order: Pholidota**
 - **Scientific Name: *Manis crassicaudata***
- **IUCN Status:** Endangered (EN).
- **Habitat in Maharashtra:** Forested tracts of the Western Ghats, Konkan belt, and Vidarbha.
- **Characteristics:** A highly unique mammal covered entirely in large, overlapping keratin scales. It has a long tubular snout with no teeth, and strong curved claws for digging. It feeds exclusively on ants and termites, acting as an essential natural pest controller for forests.
- **Primary Threats:** It holds the tragic title of being the most heavily trafficked wild mammal globally. It is poached aggressively in Maharashtra for its scales and meat, driven by baseless medicinal myths in illegal international black markets.

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(Action: Print and paste the image file **7726.png** here. It perfectly illustrates the physical adaptations of the Indian Wild Dog for pack hunting).



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3. Indian Wild Dog (Dhole)

- **Taxonomic Classification:**

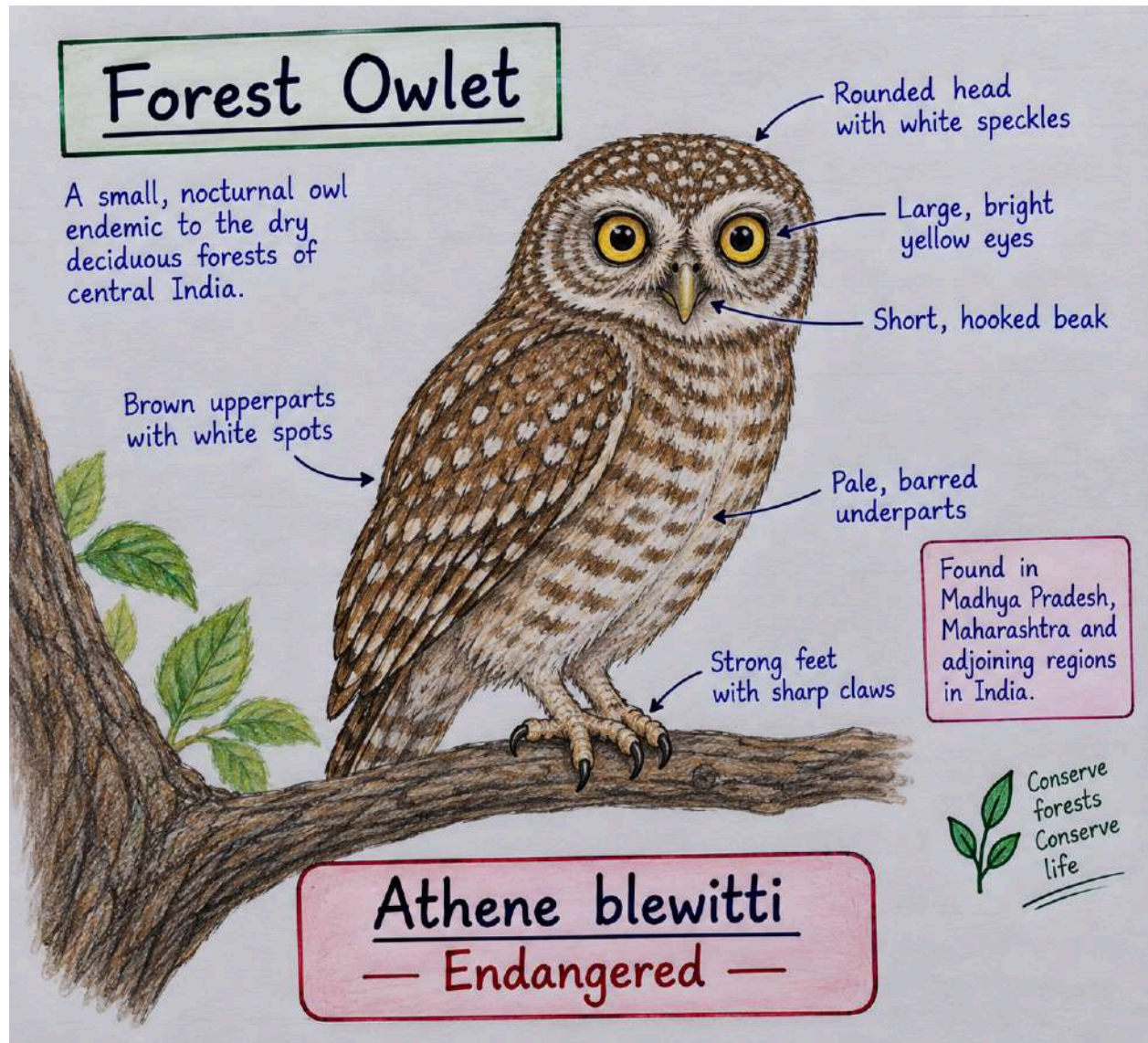
- **Kingdom: Animalia**
- **Phylum: Chordata**
- **Class: Mammalia**
- **Order: Carnivora**
- **Scientific Name: *Cuon alpinus***

- **IUCN Status:** Endangered (EN).
- **Habitat in Maharashtra:** Tadoba-Andhari Tiger Reserve, Pench National Park, and Sahyadri Tiger Reserve.
- **Characteristics:** Highly social apex predators with a reddish-brown coat and a bushy tail with a black tip. They have a lean and strong body adapted for hunting in large, coordinated packs to exhaust their prey. They regulate herbivore populations effectively.
- **Primary Threats:** Depletion of their natural prey base (like deer and wild boar) due to human hunting. They also face a severe, unique threat from fatal diseases, such as

canine distemper and rabies, transmitted by feral domestic dogs interacting with wildlife in forest buffers.

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(Action: Print and paste the image file 7727.png here. It highlights the Forest Owllet's large bright yellow eyes and specific plumage).



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4. Forest Owllet

- Taxonomic Classification:

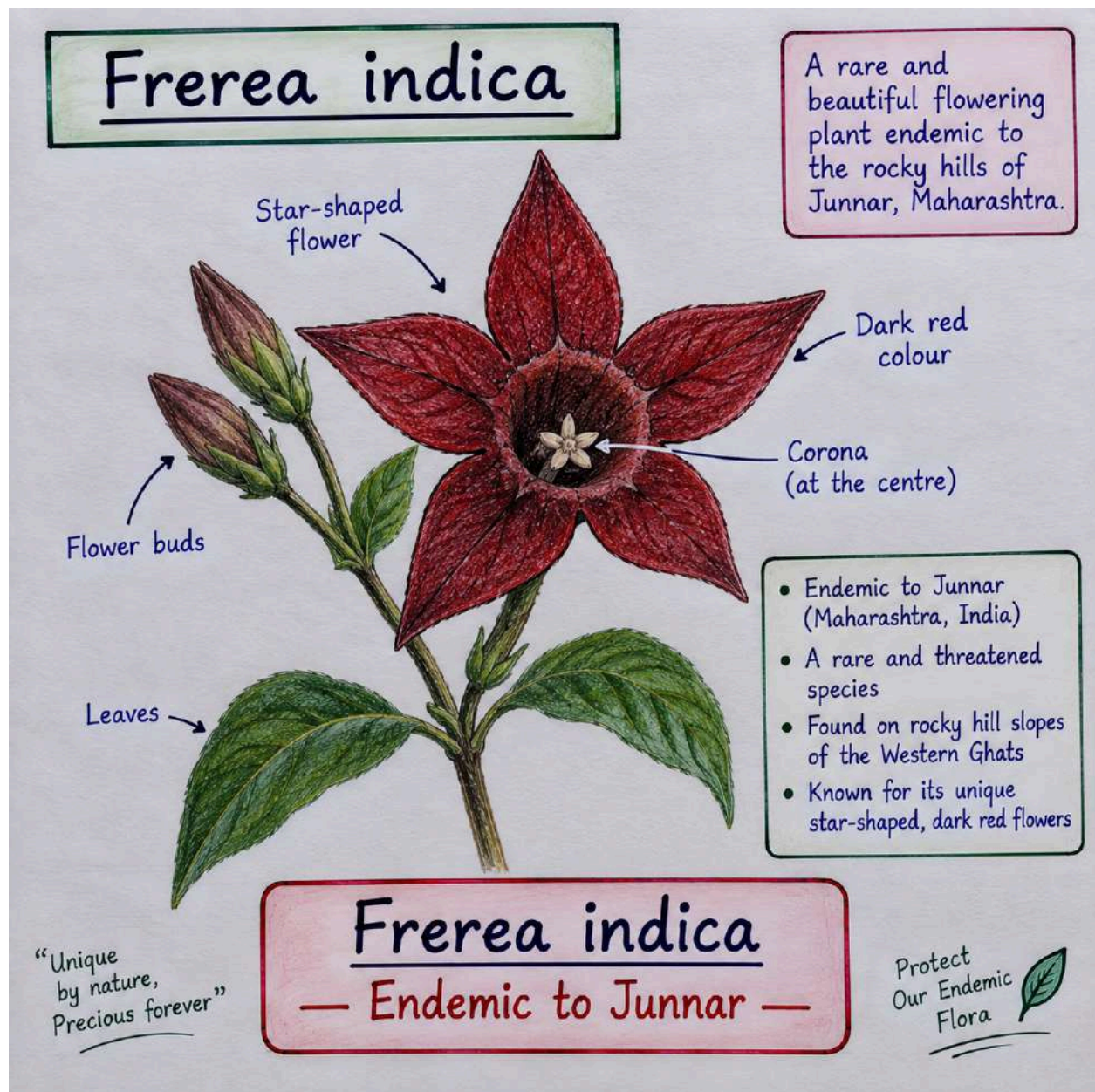
- Kingdom: Animalia
- Phylum: Chordata
- Class: Aves

- **Order: Strigiformes**
- **Scientific Name: *Athene blewitti***

- **IUCN Status:** Endangered (EN).
- **Habitat in Maharashtra:** Dry deciduous forests of the Melghat Tiger Reserve and Satpura ranges.
- **Characteristics:** A rare, small, nocturnal owl that is strictly endemic to the central Indian forests. It features a rounded head with white speckles, large bright yellow eyes, and brown upperparts. Remarkably, it was presumed extinct for 113 years before its rediscovery in 1997.
- **Primary Threats:** Rampant illegal timber logging. These owlets strictly require large, old trees with natural cavities for nesting, which are rapidly lost due to encroachment, selective logging, and human-induced forest fires.

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(Action: Print and paste the image file 7728.png here. It shows the detailed structure of the Frerea indica flower, including the corona).



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Part II: Data collected (Flora - Endangered Plants)

While animals often receive the most attention, plant species form the biological foundation of the ecosystem and are equally threatened by rapid environmental degradation.

1. Shindal Makudi

- **Taxonomic Classification:**
 - Kingdom: Plantae
 - Clade: Angiosperms
 - Order: Gentianales
 - Scientific Name: *Frereea indica*
- **IUCN Status:** Endangered (EN).

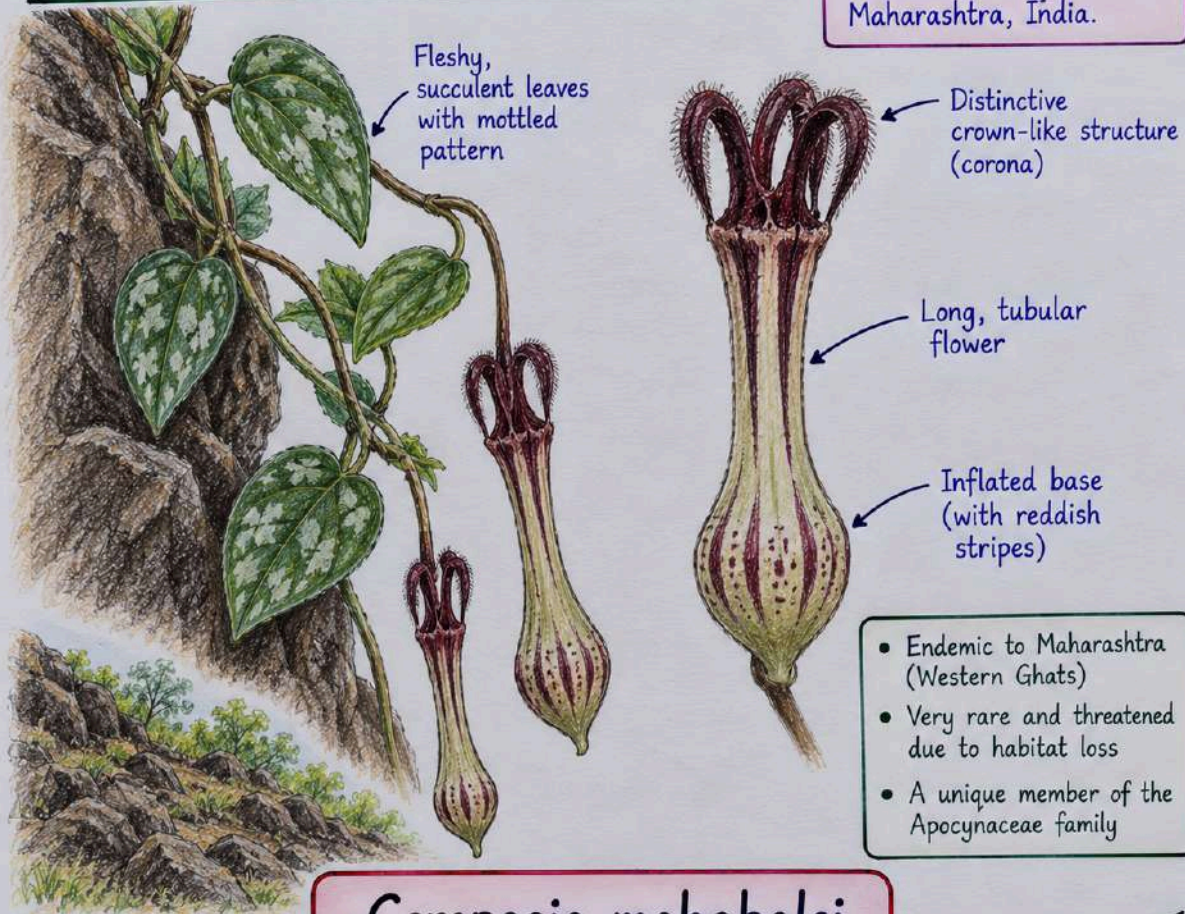
- **Habitat in Maharashtra:** Strictly endemic to the specific rocky hill slopes of Junnar in the Pune district.
- **Characteristics:** A beautiful botanical rarity globally. It is known for its unique, dark red, star-shaped flowers with a prominent corona at the center, flanked by distinct flower buds and green leaves.
- **Primary Threats:** It has a highly restricted micro-habitat. Destabilization of these cliffs due to highway road widening, localized landslides during heavy monsoons, and overgrazing by local livestock are pushing it to the brink of extinction.

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(Action: Print and paste the image file 7729.png here. It displays the fascinating tubular structure and mottled leaves of Ceropegia mahabalei).

Ceropegia mahabalei

A rare and extraordinary endemic plant species from the rocky hills of Maharashtra, India.



Natural habitat :
Rocky slopes of Western Ghats
(Maharashtra, India)

- Endemic to Maharashtra (Western Ghats)
- Very rare and threatened due to habitat loss
- A unique member of the Apocynaceae family

Ceropegia mahabalei

— Critically Endangered —

Conserve
Our Rare Flora
Conserve Biodiversity

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2. Mahabalei's Ceropegia:

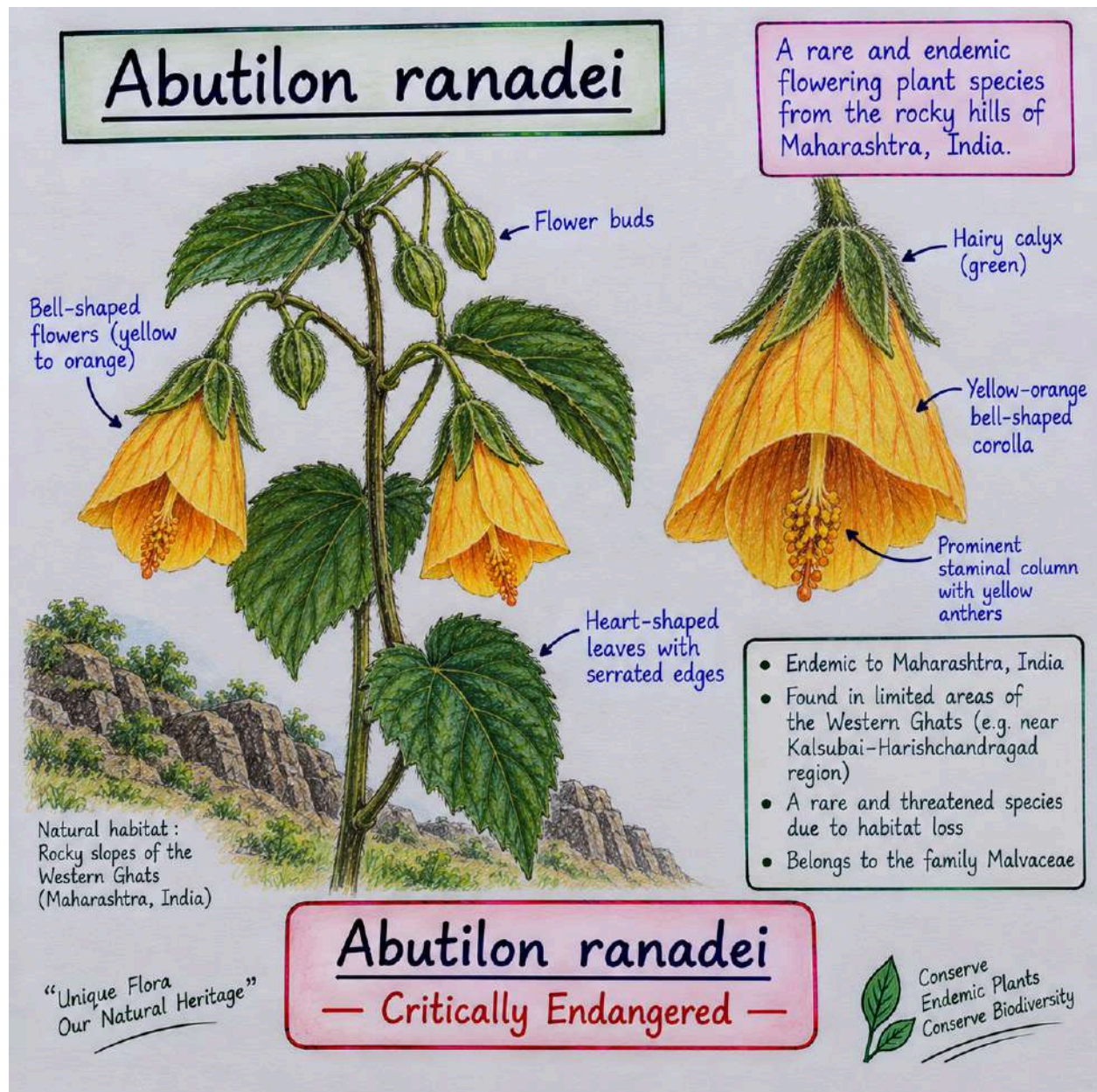
- **Taxonomic Classification:**
 - Kingdom: Plantae
 - Clade: Angiosperms
 - Order: Gentianales
 - Scientific Name: *Ceropegia mahabalei*
- **IUCN Status:** Critically Endangered (CR).
- **Habitat in Maharashtra:** Inaccessible, rocky slopes of the Sahyadri mountain ranges in the Western Ghats.
- **Characteristics:** A rare and extraordinary endemic plant featuring fleshy, succulent

leaves with a mottled pattern. It produces long, tubular flowers with an inflated base and a distinctive crown-like structure (corona) biologically adapted to temporarily trap specific insects for cross-pollination.

- **Primary Threats:** The starchy tubers of this plant are considered edible. They are heavily and unsustainably uprooted by local populations before the plant can mature and set seeds for the next generation, severely threatening its survival due to habitat loss.

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(Action: Print and paste the image file 7730.png here. It perfectly illustrates the beautiful yellow-orange bell-shaped corolla of *Abutilon ranadei*).



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3. Son Ghanta

- **Taxonomic Classification:**
 - Kingdom: Plantae
 - Clade: Angiosperms
 - Order: Malvales
 - Family: Malvaceae
 - Scientific Name: *Abutilon ranadei*
- **IUCN Status:** Critically Endangered (CR).
- **Habitat in Maharashtra:** Found in limited areas of the rocky slopes of the Western Ghats (e.g., near the Kalsubai-Harishchandragad region).
- **Characteristics:** A striking, tall endemic shrub that produces beautiful, pendulous, yellow-orange bell-shaped flowers (hence the local name 'Son Ghanta'). It has heart-shaped leaves with serrated edges and a prominent staminal column with yellow anthers.
- **Primary Threats:** Its incredibly narrow geographical distribution makes it an extremely rare and threatened species due to habitat loss. Increased eco-tourism and forest clearing for infrastructure have devastated its only known natural environment.

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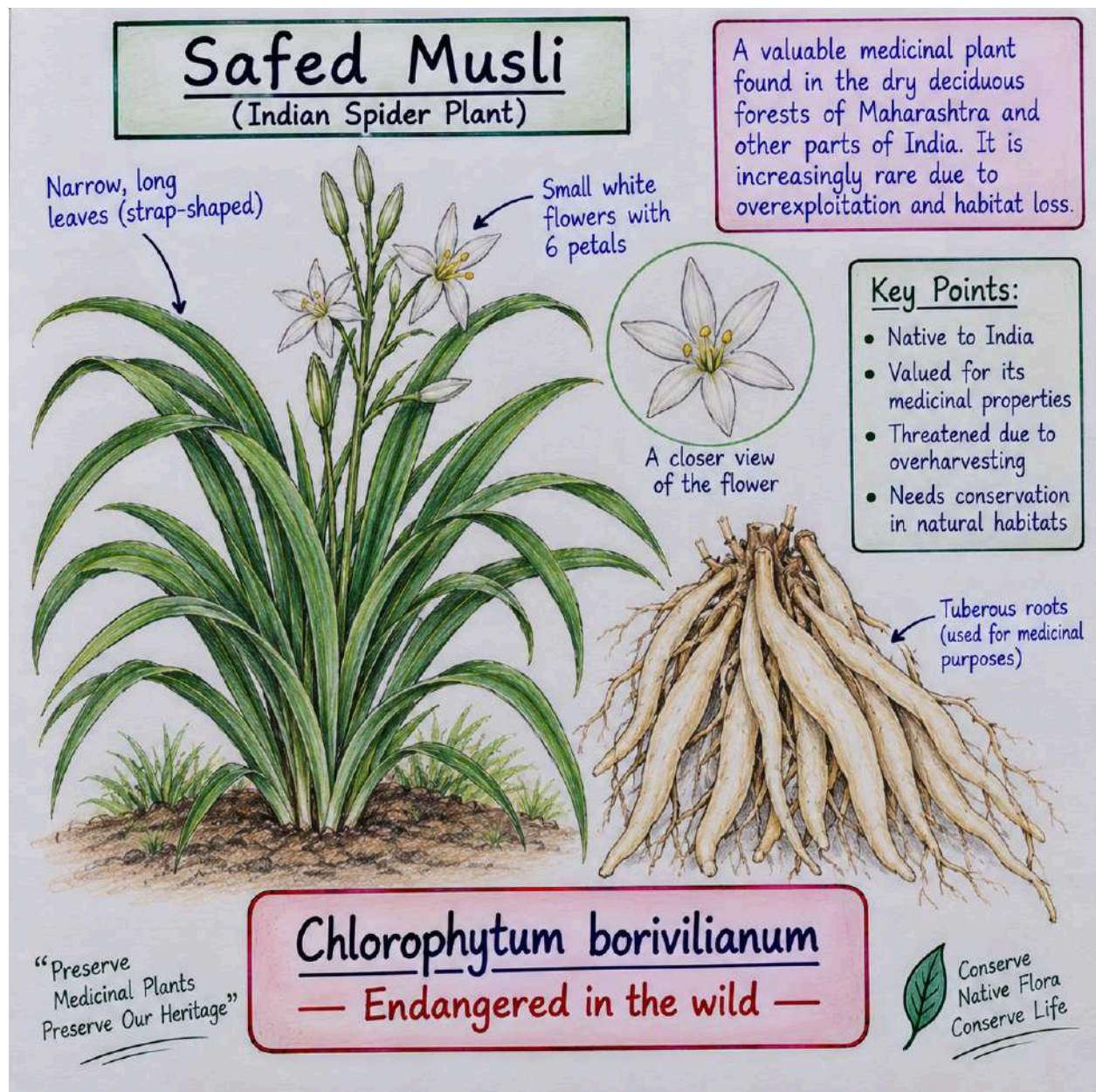
(Action: Draw a neat tabular summary of your data).

Summary of Endangered Species in Maharashtra:

Species Name	Scientific Name	Type	IUCN Status	Primary Threat
Great Indian Bustard	<i>Ardeotis nigriceps</i>	Fauna	CR	Power line collisions
Indian Pangolin	<i>Manis crassicaudata</i>	Fauna	EN	Poaching for scales
Indian Wild Dog	<i>Cuon alpinus</i>	Fauna	EN	Prey depletion & disease
Forest Owlet	<i>Athene blewitti</i>	Fauna	EN	Illegal logging
Shindal Makudi	<i>Frerea indica</i>	Flora	EN	Micro-habitat destruction
Mahabal's Ceropogia	<i>Ceropegia mahabalei</i>	Flora	CR	Unsustainable harvesting
Son Ghanta	<i>Abutilon ranadei</i>	Flora	CR	Habitat loss via tourism

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4. Safed Musli



- **Taxonomic Classification:**
 - Kingdom: Plantae
 - Clade: Angiosperms
 - Order: Asparagales
 - Scientific Name: *Chlorophytum borivilium*
- **IUCN Status:** Endangered (in the wild).
- **Habitat in Maharashtra:** Forest borders in the Western Ghats and the northern deciduous forests.
- **Characteristics:** A perennial herb deeply valued in traditional medicine for the adaptogenic properties found in its tuberous roots.
- **Primary Threats:** Rampant, unregulated uprooting of the wild plant by commercial pharmaceutical suppliers. Extracting the root completely kills the plant, preventing natural propagation and depleting the wild genetic stock.

d. Analysis of collected data

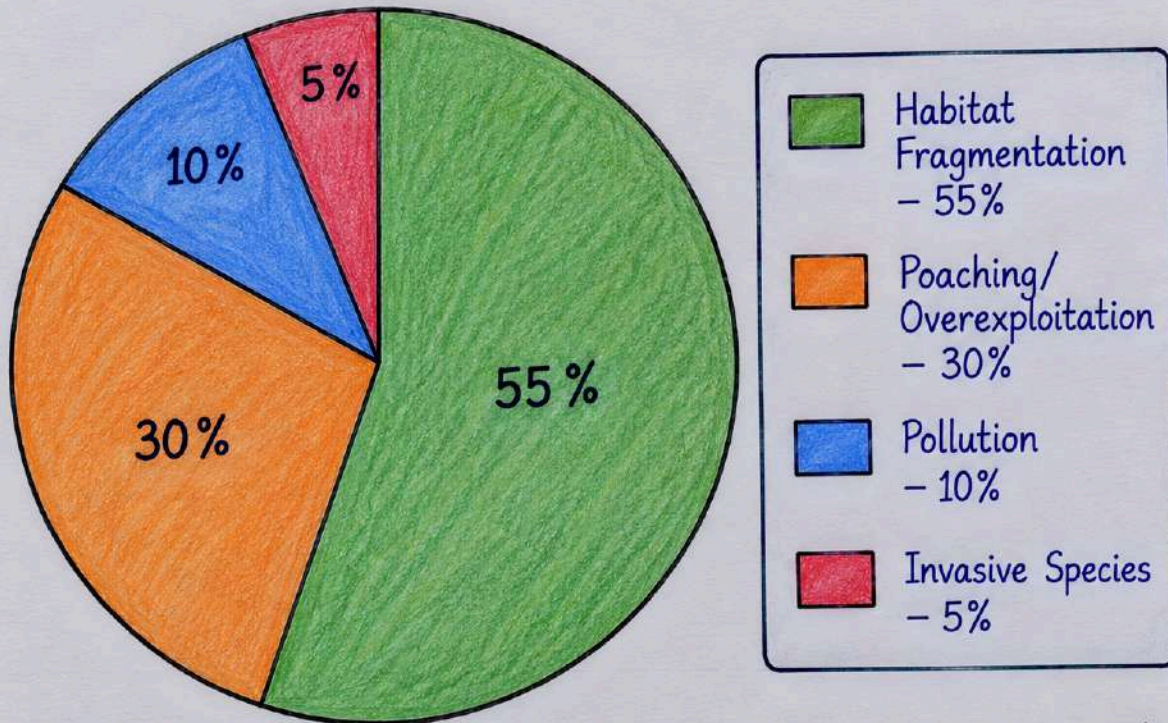
Through the formulation of the gathered information, a clear and alarming pattern of biodiversity loss emerges in Maharashtra.

1. The Dominance of Anthropogenic Stress: The data definitively proves that human (anthropogenic) interference is the overwhelming driver of endangerment. Natural causes of extinction are negligible. Whether it is physical infrastructural hazards (power lines killing Bustards) or direct, unsustainable exploitation (poaching Pangolins and over-harvesting *Ceropegia* tubers), human actions are the absolute root cause of this decline.

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(Action: Draw a Pie Chart titled "Major Causes of Extinction in Maharashtra": Habitat Fragmentation & Loss (55%), Poaching & Overexploitation (30%), Pollution & Infrastructural Hazards (10%), Invasive Species & Disease (5%).)

Major Causes of Extinction



Loss of habitats is the biggest threat to biodiversity, followed by overexploitation, pollution and invasive species.



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2. The Crisis of Habitat Fragmentation: The Western Ghats, the primary sanctuary for endemic species, is facing severe fragmentation. When a continuous forest is dissected by highways or agricultural lands, apex predators like the Dhole lose their vast hunting territories, leading to isolated populations and inbreeding. Similarly, for endemic plants like *Abutilon ranadei*, fragmentation alters the critical micro-climate—increasing temperature and decreasing humidity at forest edges—making survival impossible.

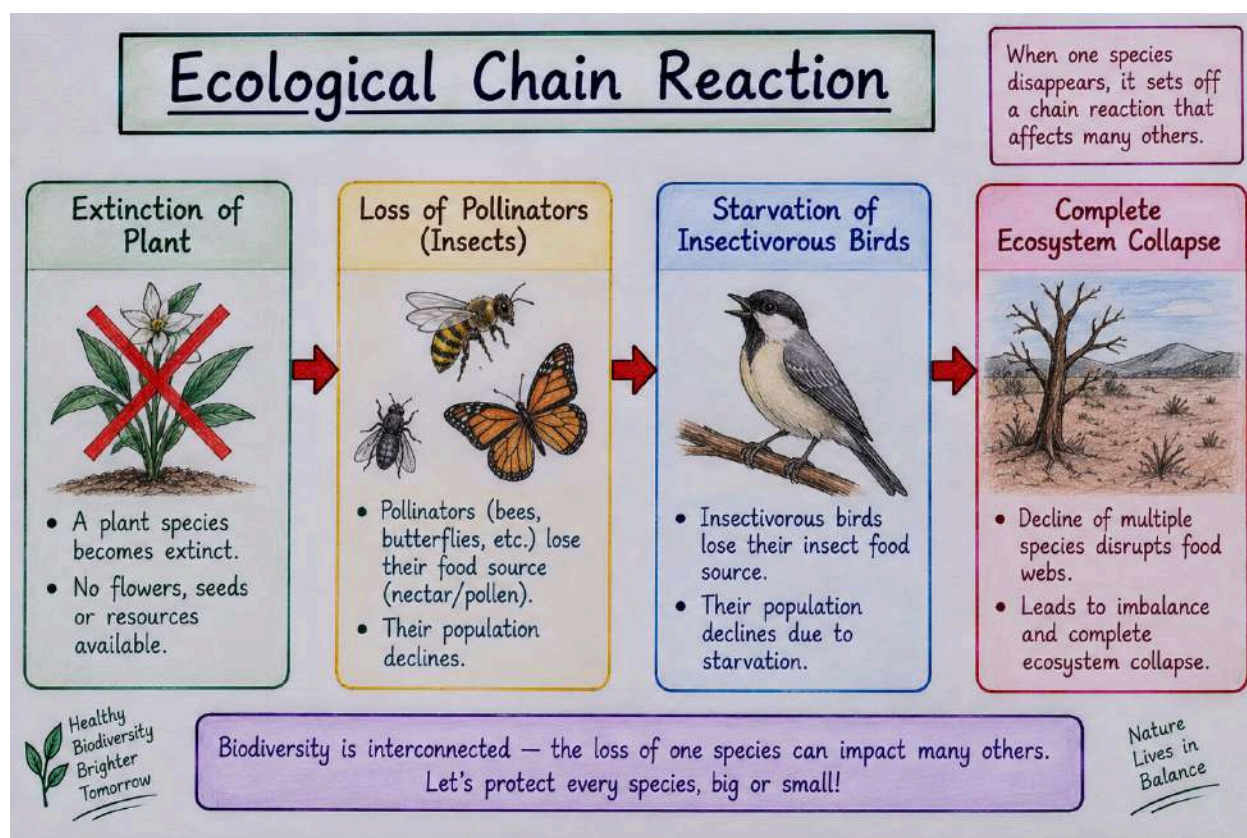
3. The Vulnerability of Endemism: The compiled data highlights that highly endemic flora and fauna are at the maximum risk. Plants like *Frerea indica* exist only on a single hill range on Earth. A single localized ecological disaster, such as a large landslide or a new mining project, could cause the immediate global extinction of the entire species, wiping out millions of years of unique evolution.

4. Cascading Ecological Impacts: The analysis proves that the removal of a single species

triggers a massive ecological chain reaction. If the Indian Pangolin is hunted to local extinction, termite populations will explode, damaging old-growth trees.

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(Action: Draw a diagram showing an "Ecological Chain Reaction": Extinction of Endemic Flora → Loss of Pollinators (Specific Insects) → Starvation of Insectivorous Birds → Complete Ecosystem Imbalance).



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Similarly, the loss of specialized endemic flora like *Ceropegia mahabalei* directly starves the highly adapted insects and birds that rely on them for nectar, collapsing the local food web.

Conservation Strategies Based on Analysis: To combat this, targeted conservation strategies must be amplified in Maharashtra:

- **In-situ Conservation:** Expanding Wildlife Sanctuaries and Tiger Reserves (like Tadoba and Melghat) to protect umbrella species, which indirectly protects the entire habitat.
- **Ex-situ Conservation:** Utilizing Botanical Gardens to cultivate rare plants like *Frerea indica* to prevent total genetic loss.
- **Devarais (Sacred Groves):** Supporting traditional practices in the Konkan where local communities protect untouched forest patches due to religious beliefs.

e. Conclusion Arriving at a conclusion based on the analysis of data, it is evident that the

biodiversity of Maharashtra is a complex and deeply fragile ecosystem. This project successfully fulfills its primary objective of identifying critically endangered flora and fauna, conclusively proving they are under immense anthropogenic stress.

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(Action: Write a bold, prominent quote in the center of the page: "The continued existence of wildlife and wilderness is important to the quality of life of humans." - Jim Fowler).

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The loss of these plant and animal species is not just a biological tragedy; it is a severe degradation of the environment that sustains human agriculture, climate regulation, and overall health.

Effective conservation requires the strict, uncompromising enforcement of the **Wildlife Protection Act (1972)** to halt poaching syndicates. Furthermore, mandatory and transparent environmental impact assessments are non-negotiable before approving any infrastructure development in the eco-sensitive zones of the Western Ghats and central Indian forests. Ultimately, the survival of Maharashtra's endangered species relies heavily on educating the youth and integrating local communities into conservation efforts. We must transition our society from a mindset of indiscriminately exploiting nature to one of consciously protecting it, ensuring our rich ecological heritage remains intact for future generations.

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f. Bibliography- references for all the information sighted in the project

Textbooks and Academic Resources:

1. State Board Biology Textbook for Class 12, Maharashtra State Bureau of Textbook Production and Curriculum Research, Pune. (Referenced for core concepts of Biodiversity and Conservation).
2. NCERT Biology Class 12 Textbook (Used for foundational ecological frameworks).

Government & Scientific Reports:

3. "Endemic and Threatened Flora of Maharashtra" – Research publications and survey data by the Botanical Survey of India (BSI), Western Regional Centre, Pune.
4. Annual Wildlife Census Reports and Action Plans – Published by the Maharashtra State Forest Department.
5. Records and Species Factsheets of the Zoological Survey of India (ZSI), Western Regional Centre, Pune.

Online Scientific Databases:

6. The IUCN Red List of Threatened Species Database (iucnredlist.org) - Accessed for confirming the exact threat status (CR, EN) of all listed flora and fauna.
7. ENVIS (Environmental Information System) Resource Partner on Avian Ecology.